

## Tables of the Binomial Cumulative Distribution

The table below gives the probability of obtaining at most  $x$  successes in  $n$  independent trials, each of which has a probability  $p$  of success. That is, if  $X$  denotes the number of successes, the table shows

$$P(X \leq x) = \sum_{r=0}^x C_r^n p^r (1-p)^{n-r}$$

| $p =$ |       | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   | 0.1    | 0.15   | 0.2    | 0.25   | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |
|-------|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $n=2$ | $x=0$ | 0.9801 | 0.9604 | 0.9409 | 0.9216 | 0.9025 | 0.8836 | 0.8649 | 0.8464 | 0.8281 | 0.8100 | 0.7225 | 0.6400 | 0.5625 | 0.4900 | 0.4225 | 0.3600 | 0.3025 | 0.2500 |
|       | 1     | 0.9999 | 0.9996 | 0.9991 | 0.9984 | 0.9975 | 0.9964 | 0.9951 | 0.9936 | 0.9919 | 0.9900 | 0.9775 | 0.9600 | 0.9375 | 0.9100 | 0.8775 | 0.8400 | 0.7975 | 0.7500 |
|       | 2     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| $n=3$ | $x=0$ | 0.9703 | 0.9412 | 0.9127 | 0.8847 | 0.8574 | 0.8306 | 0.8044 | 0.7787 | 0.7536 | 0.7290 | 0.6141 | 0.5120 | 0.4219 | 0.3430 | 0.2746 | 0.2160 | 0.1664 | 0.1250 |
|       | 1     | 0.9997 | 0.9988 | 0.9974 | 0.9953 | 0.9928 | 0.9896 | 0.9860 | 0.9818 | 0.9772 | 0.9720 | 0.9393 | 0.8960 | 0.8438 | 0.7840 | 0.7183 | 0.6480 | 0.5748 | 0.5000 |
|       | 2     | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9999 | 0.9998 | 0.9997 | 0.9995 | 0.9993 | 0.9990 | 0.9966 | 0.9920 | 0.9844 | 0.9730 | 0.9571 | 0.9360 | 0.9089 | 0.8750 |
|       | 3     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| $n=4$ | $x=0$ | 0.9606 | 0.9224 | 0.8853 | 0.8493 | 0.8145 | 0.7807 | 0.7481 | 0.7164 | 0.6857 | 0.6561 | 0.5220 | 0.4096 | 0.3164 | 0.2401 | 0.1785 | 0.1296 | 0.0915 | 0.0625 |
|       | 1     | 0.9994 | 0.9977 | 0.9948 | 0.9909 | 0.9860 | 0.9801 | 0.9733 | 0.9656 | 0.9570 | 0.9477 | 0.8905 | 0.8192 | 0.7383 | 0.6517 | 0.5630 | 0.4752 | 0.3910 | 0.3125 |
|       | 2     | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9995 | 0.9992 | 0.9987 | 0.9981 | 0.9973 | 0.9963 | 0.9880 | 0.9728 | 0.9492 | 0.9163 | 0.8735 | 0.8208 | 0.7585 | 0.6875 |
|       | 3     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9999 | 0.9995 | 0.9984 | 0.9961 | 0.9919 | 0.9850 | 0.9744 | 0.9590 | 0.9375 |
|       | 4     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| $n=5$ | $x=0$ | 0.9510 | 0.9039 | 0.8587 | 0.8154 | 0.7738 | 0.7339 | 0.6957 | 0.6591 | 0.6240 | 0.5905 | 0.4437 | 0.3277 | 0.2373 | 0.1681 | 0.1160 | 0.0778 | 0.0503 | 0.0313 |
|       | 1     | 0.9990 | 0.9962 | 0.9915 | 0.9852 | 0.9774 | 0.9681 | 0.9575 | 0.9456 | 0.9326 | 0.9185 | 0.8352 | 0.7373 | 0.6328 | 0.5282 | 0.4284 | 0.3370 | 0.2562 | 0.1875 |
|       | 2     | 1.0000 | 0.9999 | 0.9997 | 0.9994 | 0.9988 | 0.9980 | 0.9969 | 0.9955 | 0.9937 | 0.9914 | 0.9734 | 0.9421 | 0.8965 | 0.8369 | 0.7648 | 0.6826 | 0.5931 | 0.5000 |
|       | 3     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9999 | 0.9998 | 0.9997 | 0.9995 | 0.9978 | 0.9933 | 0.9844 | 0.9692 | 0.9460 | 0.9130 | 0.8688 | 0.8125 |
|       | 4     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9997 | 0.9990 | 0.9976 | 0.9947 | 0.9898 | 0.9815 | 0.9688 |
|       | 5     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| $n=6$ | $x=0$ | 0.9415 | 0.8858 | 0.8330 | 0.7828 | 0.7351 | 0.6899 | 0.6470 | 0.6064 | 0.5679 | 0.5314 | 0.3771 | 0.2621 | 0.1780 | 0.1176 | 0.0754 | 0.0467 | 0.0277 | 0.0156 |
|       | 1     | 0.9985 | 0.9943 | 0.9875 | 0.9784 | 0.9672 | 0.9541 | 0.9392 | 0.9227 | 0.9048 | 0.8857 | 0.7765 | 0.6554 | 0.5339 | 0.4202 | 0.3191 | 0.2333 | 0.1636 | 0.1094 |
|       | 2     | 1.0000 | 0.9998 | 0.9995 | 0.9988 | 0.9978 | 0.9962 | 0.9942 | 0.9915 | 0.9882 | 0.9842 | 0.9527 | 0.9011 | 0.8306 | 0.7443 | 0.6471 | 0.5443 | 0.4415 | 0.3438 |
|       | 3     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9997 | 0.9995 | 0.9992 | 0.9987 | 0.9941 | 0.9830 | 0.9624 | 0.9295 | 0.8826 | 0.8208 | 0.7447 | 0.6563 |
|       | 4     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9996 | 0.9984 | 0.9954 | 0.9891 | 0.9777 | 0.9590 | 0.9308 | 0.8906 |
|       | 5     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9993 | 0.9982 | 0.9959 | 0.9917 | 0.9844 |
|       | 6     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |





| $p=$             | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   | 0.1    | 0.15   | 0.2    | 0.25   | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |
|------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $n=14 \quad x=0$ | 0.8687 | 0.7536 | 0.6528 | 0.5647 | 0.4877 | 0.4205 | 0.3620 | 0.3112 | 0.2670 | 0.2288 | 0.1028 | 0.0440 | 0.0178 | 0.0068 | 0.0024 | 0.0008 | 0.0002 | 0.0001 |
| 1                | 0.9916 | 0.9690 | 0.9355 | 0.8941 | 0.8470 | 0.7963 | 0.7436 | 0.6900 | 0.6368 | 0.5846 | 0.3567 | 0.1979 | 0.1010 | 0.0475 | 0.0205 | 0.0081 | 0.0029 | 0.0009 |
| 2                | 0.9997 | 0.9975 | 0.9923 | 0.9833 | 0.9699 | 0.9522 | 0.9302 | 0.9042 | 0.8745 | 0.8416 | 0.6479 | 0.4481 | 0.2811 | 0.1608 | 0.0839 | 0.0398 | 0.0170 | 0.0065 |
| 3                | 1.0000 | 0.9999 | 0.9994 | 0.9981 | 0.9958 | 0.9920 | 0.9864 | 0.9786 | 0.9685 | 0.9559 | 0.8535 | 0.6982 | 0.5213 | 0.3552 | 0.2205 | 0.1243 | 0.0632 | 0.0287 |
| 4                | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9996 | 0.9990 | 0.9980 | 0.9965 | 0.9941 | 0.9908 | 0.9533 | 0.8702 | 0.7415 | 0.5842 | 0.4227 | 0.2793 | 0.1672 | 0.0898 |
| 5                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9996 | 0.9992 | 0.9985 | 0.9885 | 0.9561 | 0.8883 | 0.7805 | 0.6405 | 0.4859 | 0.3373 | 0.2120 |
| 6                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9978 | 0.9884 | 0.9617 | 0.9067 | 0.8164 | 0.6925 | 0.5461 | 0.3953 |
| 7                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9976 | 0.9897 | 0.9685 | 0.9247 | 0.8499 | 0.7414 | 0.6047 |
| 8                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9978 | 0.9917 | 0.9757 | 0.9417 | 0.8811 | 0.7880 |
| 9                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9983 | 0.9940 | 0.9825 | 0.9574 | 0.9102 |
| 10               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9989 | 0.9961 | 0.9886 | 0.9713 |
| 11               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9994 | 0.9978 | 0.9935 |
| 12               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9997 | 0.9991 |
| 13               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 |
| 14               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| $n=15 \quad x=0$ | 0.8601 | 0.7386 | 0.6333 | 0.5421 | 0.4633 | 0.3953 | 0.3367 | 0.2863 | 0.2430 | 0.2059 | 0.0874 | 0.0352 | 0.0134 | 0.0047 | 0.0016 | 0.0005 | 0.0001 | 0.0000 |
| 1                | 0.9904 | 0.9647 | 0.9270 | 0.8809 | 0.8290 | 0.7738 | 0.7168 | 0.6597 | 0.6035 | 0.5490 | 0.3186 | 0.1671 | 0.0802 | 0.0353 | 0.0142 | 0.0052 | 0.0017 | 0.0005 |
| 2                | 0.9996 | 0.9970 | 0.9906 | 0.9797 | 0.9638 | 0.9429 | 0.9171 | 0.8870 | 0.8531 | 0.8159 | 0.6042 | 0.3980 | 0.2361 | 0.1268 | 0.0617 | 0.0271 | 0.0107 | 0.0037 |
| 3                | 1.0000 | 0.9998 | 0.9992 | 0.9976 | 0.9945 | 0.9896 | 0.9825 | 0.9727 | 0.9601 | 0.9444 | 0.8227 | 0.6482 | 0.4613 | 0.2969 | 0.1727 | 0.0905 | 0.0424 | 0.0176 |
| 4                | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9994 | 0.9986 | 0.9972 | 0.9950 | 0.9918 | 0.9873 | 0.9383 | 0.8358 | 0.6865 | 0.5155 | 0.3519 | 0.2173 | 0.1204 | 0.0592 |
| 5                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9999 | 0.9997 | 0.9993 | 0.9987 | 0.9978 | 0.9832 | 0.9389 | 0.8516 | 0.7216 | 0.5643 | 0.4032 | 0.2608 | 0.1509 |
| 6                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9997 | 0.9964 | 0.9819 | 0.9434 | 0.8689 | 0.7548 | 0.6098 | 0.4522 | 0.3036 |
| 7                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9994 | 0.9958 | 0.9827 | 0.9500 | 0.8868 | 0.7869 | 0.6535 | 0.5000 |
| 8                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9992 | 0.9958 | 0.9848 | 0.9578 | 0.9050 | 0.8182 | 0.6964 |
| 9                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9992 | 0.9963 | 0.9876 | 0.9662 | 0.9231 | 0.8491 |
| 10               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9993 | 0.9972 | 0.9907 | 0.9745 | 0.9408 |
| 11               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9995 | 0.9981 | 0.9937 | 0.9824 |
| 12               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9997 | 0.9989 | 0.9963 |
| 13               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9995 |
| 14               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| 15               | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| $n=20 \quad x=0$ | 0.8179 | 0.6676 | 0.5438 | 0.4420 | 0.3585 | 0.2901 | 0.2342 | 0.1887 | 0.1516 | 0.1216 | 0.0388 | 0.0115 | 0.0032 | 0.0008 | 0.0002 | 0.0000 | 0.0000 | 0.0000 |
| 1                | 0.9831 | 0.9401 | 0.8802 | 0.8103 | 0.7358 | 0.6605 | 0.5869 | 0.5169 | 0.4516 | 0.3917 | 0.1756 | 0.0692 | 0.0243 | 0.0076 | 0.0021 | 0.0005 | 0.0001 | 0.0000 |
| 2                | 0.9990 | 0.9929 | 0.9790 | 0.9561 | 0.9245 | 0.8850 | 0.8390 | 0.7879 | 0.7334 | 0.6769 | 0.4049 | 0.2061 | 0.0913 | 0.0355 | 0.0121 | 0.0036 | 0.0009 | 0.0002 |
| 3                | 1.0000 | 0.9994 | 0.9973 | 0.9926 | 0.9841 | 0.9710 | 0.9529 | 0.9294 | 0.9007 | 0.8670 | 0.6477 | 0.4114 | 0.2252 | 0.1071 | 0.0444 | 0.0160 | 0.0049 | 0.0013 |
| 4                | 1.0000 | 1.0000 | 0.9997 | 0.9990 | 0.9974 | 0.9944 | 0.9893 | 0.9817 | 0.9710 | 0.9568 | 0.8298 | 0.6296 | 0.4148 | 0.2375 | 0.1182 | 0.0510 | 0.0189 | 0.0059 |
| 5                | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9997 | 0.9991 | 0.9981 | 0.9962 | 0.9932 | 0.9887 | 0.9327 | 0.8042 | 0.6172 | 0.4164 | 0.2454 | 0.1256 | 0.0553 | 0.0207 |
| 6                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9997 | 0.9994 | 0.9987 | 0.9976 | 0.9781 | 0.9133 | 0.7858 | 0.6080 | 0.4166 | 0.2500 | 0.1299 | 0.0577 |
| 7                | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9996 | 0.9941 | 0.9679 | 0.8982 | 0.7723 | 0.6010 | 0.4159 | 0.2520 | 0.1316 |

| $p=$         | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   | 0.1    | 0.15   | 0.2    | 0.25   | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |        |
|--------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $n=20$ $x=8$ | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9987 | 0.9900 | 0.9591 | 0.8867 | 0.7624 | 0.5956 | 0.4143 | 0.2517 |
| 9            | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9974 | 0.9861 | 0.9520 | 0.8782 | 0.7553 | 0.5914 | 0.4119 |
| 10           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9994 | 0.9961 | 0.9829 | 0.9468 | 0.8725 | 0.7507 | 0.5881 |
| 11           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9991 | 0.9949 | 0.9804 | 0.9435 | 0.8692 | 0.7483 |
| 12           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9987 | 0.9940 | 0.9790 | 0.9420 | 0.8684 |
| 13           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9985 | 0.9935 | 0.9786 | 0.9423 |        |
| 14           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9984 | 0.9936 | 0.9793 |        |
| 15           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9985 | 0.9941 |        |
| 16           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9987 |        |
| 17           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 |        |
| 18           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |        |
| $n=25$ $x=0$ | 0.7778 | 0.6035 | 0.4670 | 0.3604 | 0.2774 | 0.2129 | 0.1630 | 0.1244 | 0.0946 | 0.0718 | 0.0172 | 0.0038 | 0.0008 | 0.0001 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |        |
| 1            | 0.9742 | 0.9114 | 0.8280 | 0.7358 | 0.6424 | 0.5527 | 0.4696 | 0.3947 | 0.3286 | 0.2712 | 0.0931 | 0.0274 | 0.0070 | 0.0016 | 0.0003 | 0.0001 | 0.0000 | 0.0000 |        |
| 2            | 0.9980 | 0.9868 | 0.9620 | 0.9235 | 0.8729 | 0.8129 | 0.7466 | 0.6768 | 0.6063 | 0.5371 | 0.2537 | 0.0982 | 0.0321 | 0.0090 | 0.0021 | 0.0004 | 0.0001 | 0.0000 |        |
| 3            | 0.9999 | 0.9986 | 0.9938 | 0.9835 | 0.9659 | 0.9402 | 0.9064 | 0.8649 | 0.8169 | 0.7636 | 0.4711 | 0.2340 | 0.0962 | 0.0332 | 0.0097 | 0.0024 | 0.0005 | 0.0001 |        |
| 4            | 1.0000 | 0.9999 | 0.9992 | 0.9972 | 0.9928 | 0.9850 | 0.9726 | 0.9549 | 0.9314 | 0.9020 | 0.6821 | 0.4207 | 0.2137 | 0.0905 | 0.0320 | 0.0095 | 0.0023 | 0.0005 |        |
| 5            | 1.0000 | 1.0000 | 0.9999 | 0.9996 | 0.9988 | 0.9969 | 0.9935 | 0.9877 | 0.9790 | 0.9666 | 0.8385 | 0.6167 | 0.3783 | 0.1935 | 0.0826 | 0.0294 | 0.0086 | 0.0020 |        |
| 6            | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9995 | 0.9987 | 0.9972 | 0.9946 | 0.9905 | 0.9305 | 0.7800 | 0.5611 | 0.3407 | 0.1734 | 0.0736 | 0.0258 | 0.0073 |        |
| 7            | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9995 | 0.9989 | 0.9977 | 0.9745 | 0.8909 | 0.7265 | 0.5118 | 0.3061 | 0.1536 | 0.0639 | 0.0216 |        |
| 8            | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9995 | 0.9920 | 0.9532 | 0.8506 | 0.6769 | 0.4668 | 0.2735 | 0.1340 | 0.0539 |        |
| 9            | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9979 | 0.9827 | 0.9287 | 0.8106 | 0.6303 | 0.4246 | 0.2424 | 0.1148 |        |
| 10           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9995 | 0.9944 | 0.9703 | 0.9022 | 0.7712 | 0.5858 | 0.3843 | 0.2122 |        |
| 11           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9985 | 0.9893 | 0.9558 | 0.8746 | 0.7323 | 0.5426 | 0.3450 |        |
| 12           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9966 | 0.9825 | 0.9396 | 0.8462 | 0.6937 | 0.5000 |        |
| 13           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9991 | 0.9940 | 0.9745 | 0.9222 | 0.8173 | 0.6550 |        |
| 14           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9982 | 0.9907 | 0.9656 | 0.9040 | 0.7878 |        |
| 15           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9995 | 0.9971 | 0.9868 | 0.9560 | 0.8852 |        |
| 16           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9992 | 0.9957 | 0.9826 | 0.9461 |        |
| 17           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9988 | 0.9942 | 0.9784 |        |
| 18           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9984 | 0.9927 |        |
| 19           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9996 | 0.9980 |        |
| 20           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9995 |        |
| 21           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 |        |
| 22           | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |        |
| $n=30$ $x=0$ | 0.7397 | 0.5455 | 0.4010 | 0.2939 | 0.2146 | 0.1563 | 0.1134 | 0.0820 | 0.0591 | 0.0424 | 0.0076 | 0.0012 | 0.0002 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |        |
| 1            | 0.9639 | 0.8795 | 0.7731 | 0.6612 | 0.5535 | 0.4555 | 0.3694 | 0.2958 | 0.2343 | 0.1837 | 0.0480 | 0.0105 | 0.0020 | 0.0003 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |        |
| 2            | 0.9967 | 0.9783 | 0.9399 | 0.8831 | 0.8122 | 0.7324 | 0.6487 | 0.5654 | 0.4855 | 0.4114 | 0.1514 | 0.0442 | 0.0106 | 0.0021 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |        |
| 3            | 0.9998 | 0.9971 | 0.9881 | 0.9694 | 0.9392 | 0.8974 | 0.8450 | 0.7842 | 0.7175 | 0.6474 | 0.3217 | 0.1227 | 0.0374 | 0.0093 | 0.0019 | 0.0003 | 0.0000 | 0.0000 |        |
| 4            | 1.0000 | 0.9997 | 0.9982 | 0.9937 | 0.9844 | 0.9685 | 0.9447 | 0.9126 | 0.8723 | 0.8245 | 0.5245 | 0.2552 | 0.0979 | 0.0302 | 0.0075 | 0.0015 | 0.0002 | 0.0000 |        |

| $p=$   |        | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   | 0.1    | 0.15   | 0.2    | 0.25   | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |
|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| $n=30$ | $x=5$  | 1.0000 | 1.0000 | 0.9998 | 0.9989 | 0.9967 | 0.9921 | 0.9838 | 0.9707 | 0.9519 | 0.9268 | 0.7106 | 0.4275 | 0.2026 | 0.0766 | 0.0233 | 0.0057 | 0.0011 | 0.0002 |
|        | 6      | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9994 | 0.9983 | 0.9960 | 0.9918 | 0.9848 | 0.9742 | 0.8474 | 0.6070 | 0.3481 | 0.1595 | 0.0586 | 0.0172 | 0.0040 | 0.0007 |
|        | 7      | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9997 | 0.9992 | 0.9980 | 0.9959 | 0.9922 | 0.9302 | 0.7608 | 0.5143 | 0.2814 | 0.1238 | 0.0435 | 0.0121 | 0.0026 |
|        | 8      | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9996 | 0.9990 | 0.9980 | 0.9722 | 0.8713 | 0.6736 | 0.4315 | 0.2247 | 0.0940 | 0.0312 | 0.0081 |
|        | 9      | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9995 | 0.9903 | 0.9389 | 0.8034 | 0.5888 | 0.3575 | 0.1763 | 0.0694 | 0.0214 |
|        | 10     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9971 | 0.9744 | 0.8943 | 0.7304 | 0.5078 | 0.2915 | 0.1350 | 0.0494 |
|        | 11     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9992 | 0.9905 | 0.9493 | 0.8407 | 0.6548 | 0.4311 | 0.2327 | 0.1002 |
|        | 12     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9969 | 0.9784 | 0.9155 | 0.7802 | 0.5785 | 0.3592 | 0.1808 |
|        | 13     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9991 | 0.9918 | 0.9599 | 0.8737 | 0.7145 | 0.5025 | 0.2923 |
|        | 14     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9973 | 0.9831 | 0.9348 | 0.8246 | 0.6448 | 0.4278 |
|        | 15     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9992 | 0.9936 | 0.9699 | 0.9029 | 0.7691 | 0.5722 |
|        | 16     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9979 | 0.9876 | 0.9519 | 0.8644 | 0.7077 |
|        | 17     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9994 | 0.9955 | 0.9788 | 0.9286 | 0.8192 |
|        | 18     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9986 | 0.9917 | 0.9666 | 0.8998 |
|        | 19     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9971 | 0.9862 | 0.9506 |
|        | 20     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9991 | 0.9950 | 0.9786 |
|        | 21     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9984 | 0.9919 |
|        | 22     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9974 |
|        | 23     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9993 |
|        | 24     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 |
|        | 25     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| $n=40$ | $x=0$  | 0.6690 | 0.4457 | 0.2957 | 0.1954 | 0.1285 | 0.0842 | 0.0549 | 0.0356 | 0.0230 | 0.0148 | 0.0015 | 0.0001 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
|        | 1      | 0.9393 | 0.8095 | 0.6615 | 0.5210 | 0.3991 | 0.2990 | 0.2201 | 0.1594 | 0.1140 | 0.0805 | 0.0121 | 0.0015 | 0.0001 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
|        | 2      | 0.9925 | 0.9543 | 0.8822 | 0.7855 | 0.6767 | 0.5665 | 0.4625 | 0.3694 | 0.2894 | 0.2228 | 0.0486 | 0.0079 | 0.0010 | 0.0001 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
|        | 3      | 0.9993 | 0.9918 | 0.9686 | 0.9252 | 0.8619 | 0.7827 | 0.6937 | 0.6007 | 0.5092 | 0.4231 | 0.1302 | 0.0285 | 0.0047 | 0.0006 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
|        | 4      | 1.0000 | 0.9988 | 0.9933 | 0.9790 | 0.9520 | 0.9104 | 0.8546 | 0.7868 | 0.7103 | 0.6290 | 0.2633 | 0.0759 | 0.0160 | 0.0026 | 0.0003 | 0.0000 | 0.0000 | 0.0000 |
|        | 5      | 1.0000 | 0.9999 | 0.9988 | 0.9951 | 0.9861 | 0.9691 | 0.9419 | 0.9033 | 0.8535 | 0.7937 | 0.4325 | 0.1613 | 0.0433 | 0.0086 | 0.0013 | 0.0001 | 0.0000 | 0.0000 |
|        | 6      | 1.0000 | 1.0000 | 0.9998 | 0.9990 | 0.9966 | 0.9909 | 0.9801 | 0.9624 | 0.9361 | 0.9005 | 0.6067 | 0.2859 | 0.0962 | 0.0238 | 0.0044 | 0.0006 | 0.0001 | 0.0000 |
|        | 7      | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9993 | 0.9977 | 0.9942 | 0.9873 | 0.9758 | 0.9581 | 0.7559 | 0.4371 | 0.1820 | 0.0553 | 0.0124 | 0.0021 | 0.0002 | 0.0000 |
|        | 8      | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9995 | 0.9985 | 0.9963 | 0.9919 | 0.9845 | 0.8646 | 0.5931 | 0.2998 | 0.1110 | 0.0303 | 0.0061 | 0.0009 | 0.0001 |
|        | 9      | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9997 | 0.9990 | 0.9976 | 0.9949 | 0.9328 | 0.7318 | 0.4395 | 0.1959 | 0.0644 | 0.0156 | 0.0027 | 0.0003 |
|        | $x=10$ | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9998 | 0.9994 | 0.9985 | 0.9701 | 0.8392 | 0.5839 | 0.3087 | 0.1215 | 0.0352 | 0.0074 | 0.0011 |
|        | 11     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9996 | 0.9880 | 0.9125 | 0.7151 | 0.4406 | 0.2053 | 0.0709 | 0.0179 | 0.0032 |
|        | 12     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9957 | 0.9568 | 0.8209 | 0.5772 | 0.3143 | 0.1285 | 0.0386 | 0.0083 |
|        | 13     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9986 | 0.9806 | 0.8968 | 0.7032 | 0.4408 | 0.2112 | 0.0751 | 0.0192 |
|        | 14     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9921 | 0.9456 | 0.8074 | 0.5721 | 0.3174 | 0.1326 | 0.0403 |
|        | 15     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9971 | 0.9738 | 0.8849 | 0.6946 | 0.4402 | 0.2142 | 0.0769 |
|        | 16     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9990 | 0.9884 | 0.9367 | 0.7978 | 0.5681 | 0.3185 | 0.1341 |
|        | 17     | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9953 | 0.9680 | 0.8761 | 0.6885 | 0.4391 | 0.2148 |

| $p=$                                       | 0.01   | 0.02   | 0.03   | 0.04   | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   | 0.1    | 0.15   | 0.2    | 0.25   | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |
|--------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| <b><math>n=40</math> <math>x=18</math></b> | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9983 | 0.9852 | 0.9301 | 0.7911 | 0.5651 | 0.3179 |
| <b>19</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9994 | 0.9937 | 0.9637 | 0.8702 | 0.6844 | 0.4373 |
| <b>20</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9976 | 0.9827 | 0.9256 | 0.7870 | 0.5627 |
| <b>21</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9991 | 0.9925 | 0.9608 | 0.8669 | 0.6821 |
| <b>22</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9970 | 0.9811 | 0.9233 | 0.7852 |
| <b>23</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9989 | 0.9917 | 0.9595 | 0.8659 |
| <b>24</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9966 | 0.9804 | 0.9231 |
| <b>25</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9988 | 0.9914 | 0.9597 |
| <b>26</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9966 | 0.9808 |
| <b>27</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9988 | 0.9917 |
| <b>28</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9968 |
| <b>29</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9989 |
| <b>30</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 |
| <b>31</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 |
| <b>32</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 |
| <b><math>n=50</math> <math>x=0</math></b>  | 0.6050 | 0.3642 | 0.2181 | 0.1299 | 0.0769 | 0.0453 | 0.0266 | 0.0155 | 0.0090 | 0.0052 | 0.0003 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| <b>1</b>                                   | 0.9106 | 0.7358 | 0.5553 | 0.4005 | 0.2794 | 0.1900 | 0.1265 | 0.0827 | 0.0532 | 0.0338 | 0.0029 | 0.0002 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| <b>2</b>                                   | 0.9862 | 0.9216 | 0.8108 | 0.6767 | 0.5405 | 0.4162 | 0.3108 | 0.2260 | 0.1605 | 0.1117 | 0.0142 | 0.0013 | 0.0001 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| <b>3</b>                                   | 0.9984 | 0.9822 | 0.9372 | 0.8609 | 0.7604 | 0.6473 | 0.5327 | 0.4253 | 0.3303 | 0.2503 | 0.0460 | 0.0057 | 0.0005 | 0.0000 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| <b>4</b>                                   | 0.9999 | 0.9968 | 0.9832 | 0.9510 | 0.8964 | 0.8206 | 0.7290 | 0.6290 | 0.5277 | 0.4312 | 0.1121 | 0.0185 | 0.0021 | 0.0002 | 0.0000 | 0.0000 | 0.0000 | 0.0000 |
| <b>5</b>                                   | 1.0000 | 0.9995 | 0.9963 | 0.9856 | 0.9622 | 0.9224 | 0.8650 | 0.7919 | 0.7072 | 0.6161 | 0.2194 | 0.0480 | 0.0070 | 0.0007 | 0.0001 | 0.0000 | 0.0000 | 0.0000 |
| <b>6</b>                                   | 1.0000 | 0.9999 | 0.9993 | 0.9964 | 0.9882 | 0.9711 | 0.9417 | 0.8981 | 0.8404 | 0.7702 | 0.3613 | 0.1034 | 0.0194 | 0.0025 | 0.0002 | 0.0000 | 0.0000 | 0.0000 |
| <b>7</b>                                   | 1.0000 | 1.0000 | 0.9999 | 0.9992 | 0.9968 | 0.9906 | 0.9780 | 0.9562 | 0.9232 | 0.8779 | 0.5188 | 0.1904 | 0.0453 | 0.0073 | 0.0008 | 0.0001 | 0.0000 | 0.0000 |
| <b>8</b>                                   | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9992 | 0.9973 | 0.9927 | 0.9833 | 0.9672 | 0.9421 | 0.6681 | 0.3073 | 0.0916 | 0.0183 | 0.0025 | 0.0002 | 0.0000 | 0.0000 |
| <b>9</b>                                   | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9993 | 0.9978 | 0.9944 | 0.9875 | 0.9755 | 0.7911 | 0.4437 | 0.1637 | 0.0402 | 0.0067 | 0.0008 | 0.0001 | 0.0000 |
| <b>10</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9994 | 0.9983 | 0.9957 | 0.9906 | 0.8801 | 0.5836 | 0.2622 | 0.0789 | 0.0160 | 0.0022 | 0.0002 | 0.0000 |
| <b>11</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9995 | 0.9987 | 0.9968 | 0.9372 | 0.7107 | 0.3816 | 0.1390 | 0.0342 | 0.0057 | 0.0006 | 0.0000 |
| <b>12</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9996 | 0.9990 | 0.9699 | 0.8139 | 0.5110 | 0.2229 | 0.0661 | 0.0133 | 0.0018 | 0.0002 |
| <b>13</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9997 | 0.9868 | 0.8894 | 0.6370 | 0.3279 | 0.1163 | 0.0280 | 0.0045 | 0.0005 |
| <b>14</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9947 | 0.9393 | 0.7481 | 0.4468 | 0.1878 | 0.0540 | 0.0104 | 0.0013 |
| <b>15</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9981 | 0.9692 | 0.8369 | 0.5692 | 0.2801 | 0.0955 | 0.0220 | 0.0033 |
| <b>16</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9993 | 0.9856 | 0.9017 | 0.6839 | 0.3889 | 0.1561 | 0.0427 | 0.0077 |
| <b><math>n=50</math> <math>x=17</math></b> | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9998 | 0.9937 | 0.9449 | 0.7822 | 0.5060 | 0.2369 | 0.0765 | 0.0164 |
| <b>18</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9975 | 0.9713 | 0.8594 | 0.6216 | 0.3356 | 0.1273 | 0.0325 |
| <b>19</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9991 | 0.9861 | 0.9152 | 0.7264 | 0.4465 | 0.1974 | 0.0595 |
| <b>20</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9997 | 0.9937 | 0.9522 | 0.8139 | 0.5610 | 0.2862 | 0.1013 |
| <b>21</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9999 | 0.9974 | 0.9749 | 0.8813 | 0.6701 | 0.3900 | 0.1611 |
| <b>22</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9990 | 0.9877 | 0.9290 | 0.7660 | 0.5019 | 0.2399 |
| <b>23</b>                                  | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 1.0000 | 0.9996 | 0.9944 | 0.9604 | 0.8438 | 0.6134 | 0.3359 |

[illegible]